The Role of Memory in Noisy-Channel Inference during Language Comprehension

The goal of communication is to transmit information from a producer to a comprehender, yet comprehenders often arrive at an interpretation that differs from the literal meaning of a sentence [1]. According to the Noisy Channel theory, comprehenders infer the speaker's intended meaning from what they perceive, combining the prior probability of intended meanings with a model of likely errors [2,3]. Prior work shows that comprehenders are more likely to make an inference when implausible input may be the result of an error (e.g., deletion of "to" in "The sister mailed the letter the niece"; [4]). [5] further demonstrated that discourse context affects these inferences via the prior, with supportive contexts resulting in more inferences relative to unsupportive contexts. What remains unclear is the role of memory in these inferences.

Methods: Across 3 experiments, 534 participants recruited via Prolific (U.S.-based, approval $\geq$95%, paid $13/hr) met inclusion criteria ($\geq$75% filler accuracy). Each participant completed 71 trials (from [5]): 3 practice, 20 target (10 plausible, 10 implausible), and 48 fillers. Each trial included two context sentences, a target sentence, a comprehension question, and a prompt to type the target sentence from memory (all in English). Target sentences varied by plausibility, structure/noise type (**Table 1**), and discourse context: supportive (preceding sentences formed a coherent narrative; S) or unsupportive (preceding sentences unrelated; US). In E1, sentences appeared in RSVP (500ms/word, 150ms ISI). A Yes/No comprehension question was also presented in RSVP (500ms/word, 140ms ISI). Finally, participants were asked to type out the target sentence from memory. In E2, the sentences and question were presented all at once (but on separate screens). To equate the total time on the task, participants were only able to respond to the comprehension question after a delay yoked to the same item in E1 (total time to display stimulus + median of RT). E3 was a direct replication of [5]. All sentences and questions were shown on the screen simultaneously, and participants responded at their own pace. There were no memory prompts.

Results: *Comprehension.* For implausible sentences (**Fig 1**), PO sentences were more likely to be interpreted literally than DO sentences (E1: β = -.89, 95% CrI = [-1.13, -0.62]; E2: β = -1.04, 95% CrI = [-1.3, -.78]; E3: β = -.49, 95% CrI = [-0.71, -.26]). Active sentences had the highest rate of literal interpretation and were more likely to be interpreted literally than PO sentences (E1: β = -1.3, 95% CrI = [-1.74, -0.85]; E2: β = -1.85, 95% CrI = [-2.42 -1.32]; E3: β = -2.26, 95% CrI = [-2.73, -1.82]), as were passive sentences (E1: β = -0.85, 95% CrI = [-1.3, -0.41]; E2: β = -1.22, 95% CrI = [-1.71,-0.73]; E3: β = -1.9, 95% CrI = [-2.32,-1.47]). Supportive context increased the proportion of inferences in E3 but not in E1 or E2 (E1: $\beta = 0.17$, 95% CrI = [-.21, .54]; E2: $\beta = 0.15$, 95% CrI = [-.3, .58]; E3: $\beta = 0.40$, 95% CrI = [.01, .78]), though there was little support for an interaction between context and experiment (β = .11, 95% CrI = [-.16, .38]). *Memory.* E1 memory responses were annotated with GPT-4o, which showed high agreement with a human annotator for a subset of data (**Table 2**). Recall of DO sentences was more likely to contain an error than PO, active, or passive sentences (**Fig 2A**). When memory errors were introduced, the average Levenshtein distance was shorter for DO in comparison to PO, active, and passive (**Fig 2B**). Memory errors were classified into deletions, insertions, substitutions, and exchanges (**Fig 2C**). Insertions occurred more for DO than PO, active, and passive sentences. Deletions occurred more for PO than DO, active, and passive sentences. Substitutions and exchanges were more rare, and occurred most for DO and PO sentences respectively. Sentences that were more likely to elicit memory distortion were also more likely to elicit inference during comprehension (r = .60, 95% CrI = [.50, .69]) (**Fig 2D**).

Conclusions: We replicated prior work [4,5] showing that implausible DO sentences lead to more inferences than PO, active, or passive sentences, consistent with the idea that deletions represent more probable noise corruptions than insertions or exchanges. Recall data lend support to this proposed noise model. Supportive discourse lead to more inferences when memory demands were minimized. The effect of local discourse context on the priors may depend on the ability to maintain the context in working memory. Sentences that were more likely to be misremembered were also more likely to elicit inferences during comprehension: Noisy Channel inferences may reflect memory distortions rather than explicit corrections.

References: **1.** Ferreira & Patson (2007). *Lang. & Ling. Compass*, 1. 71-83. **2.** Shannon, C. E. (1948). *Bell System Technical Journal, 27(3)*, 379–423. **3.** Levy, R. (2008). *Proceedings of EMNLP*, 234–243. **4.** Gibson, E., Bergen, L., & Piantadosi, S. T. (2013). *PNAS*, 110(20), 8051–8056. **5.** Chen, S., Nathaniel, S., Ryskin, R., & Gibson, E. (2023). *Cognition, 238*, 105503.

Structure	Implausible Example	Possible Noise Types
Direct-Object (DO)	The sister mailed the letter the niece.	One-word deletion of "to"
Prepositional-Object (PO)	The sister mailed the niece to the letter.	One-word insertion of "to"
Active	The diamond lost the woman.	Two-word exchange of "woman" and "diamond"
Passive	The woman was lost by the diamond.	Two-word exchange of "woman" and "diamond"

Table 1: Implausible sentences examples

Human vs.GPT-4o	% Agreement
Overall Sentence Structure	99%
Meaning	88%
Plausibility	79%
Insertion	89%
Deletion	74%
Substitution	86%
Exchange	96%

Table 2: Agreement Human—GPT-4o

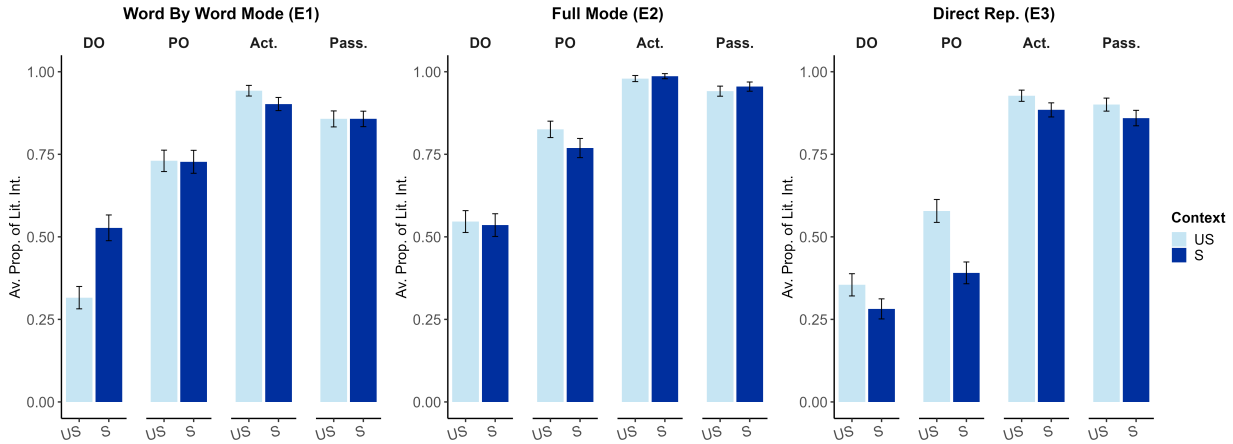

Figure 1: Proportions of literal interpretations for implausible sentences. S = Supportive discourse context. US = Unsupportive discourse context. Error bars indicate SEM.

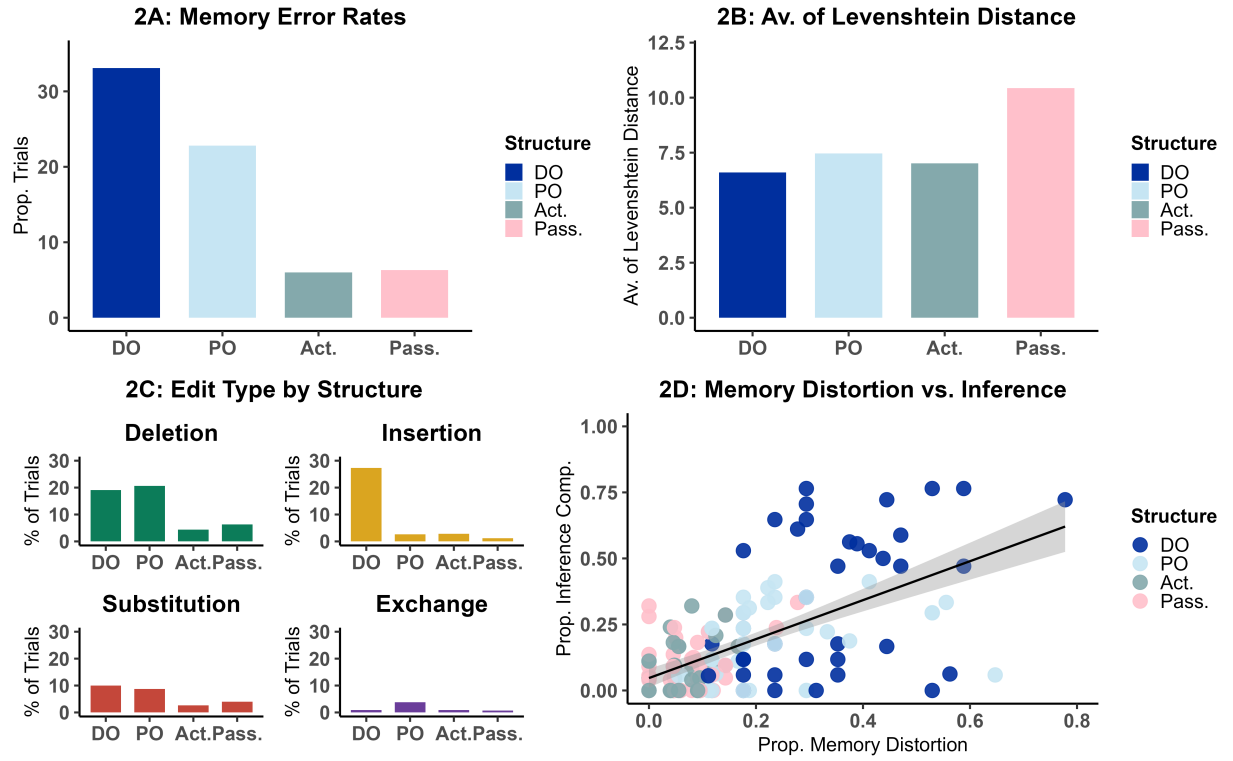

Figure 2: Memory recall results: Memory Error Rates (2A), Average Levenshtein Edit Distance (2B), Edit Types (2C), and Memory Distortion vs. Inference (2D).